

OptiSphere V4: A Compact Spherical Device for Real-Time Biological Experiments

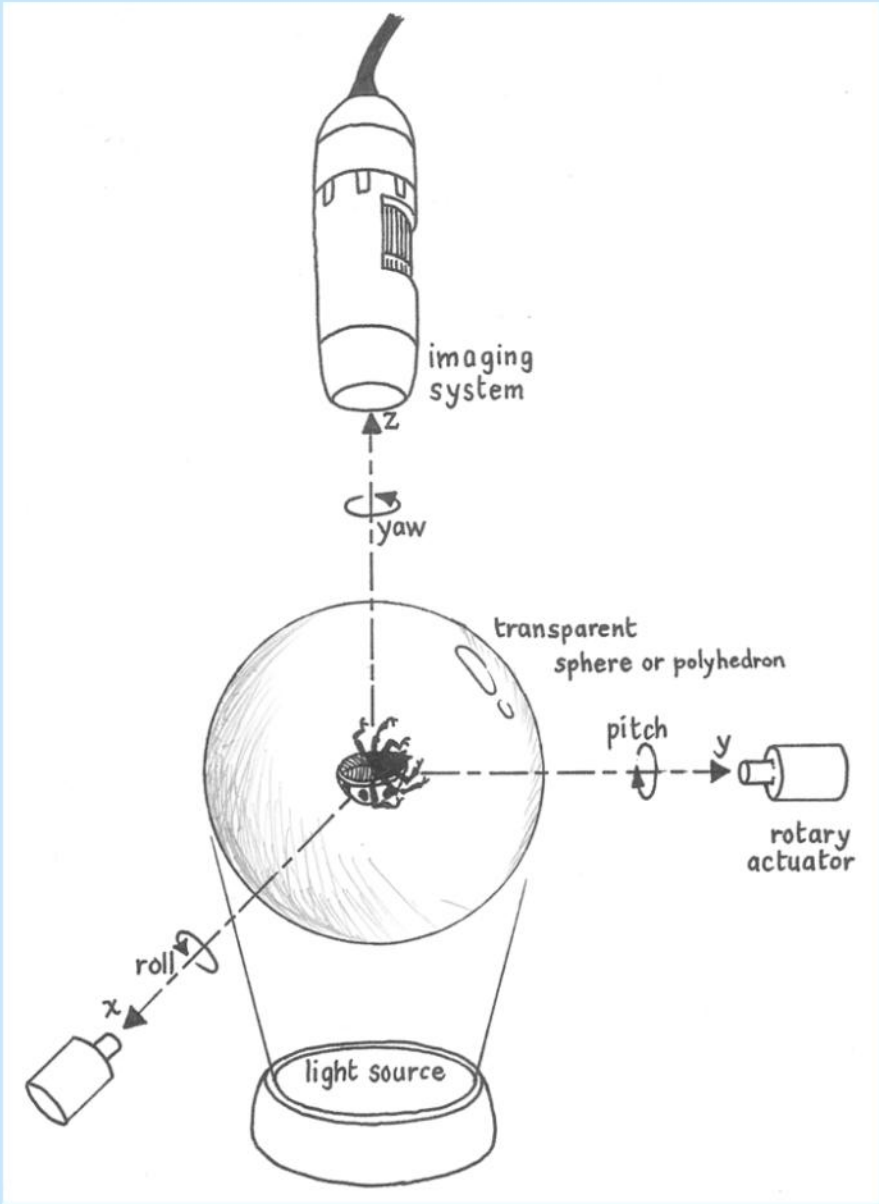
OptiSphere V4: リアルタイム生物実験用コンパクト球体デバイス

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Introduction

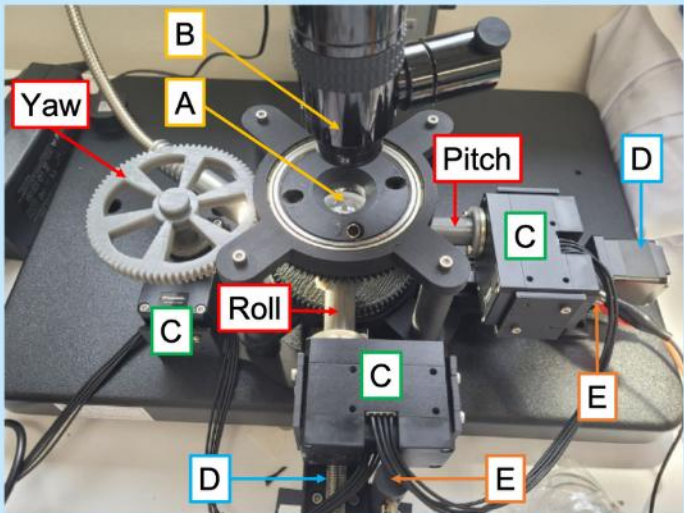
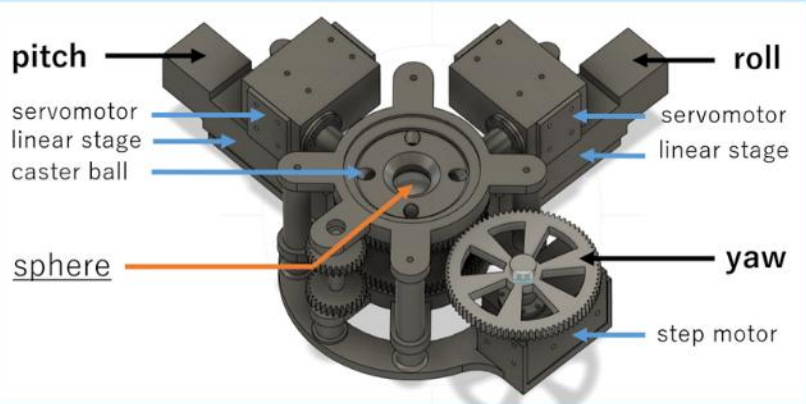
A specimen is placed either at the center or on the inner surface of a transparent sphere, with 3 rotational axis (pitch, roll, yaw), enabling multi-angle imaging but especially posture tracking of moving biological targets. Its purpose is to deliver precise, multidirectional views by combining controlled rotations and optimized mechanical design.



Galipon J. & Tadakuma K. RJST 2022 patent filed 2022-140257 patent published 2024-35660

Previous works

Past versions of the system were already developed with full electronic integration, but the system is complex, using a total of 5 motors.



- A: sphere
- B: digital microscope
- C: servomotors
- D: linear stages
- E: limit switches for calibration

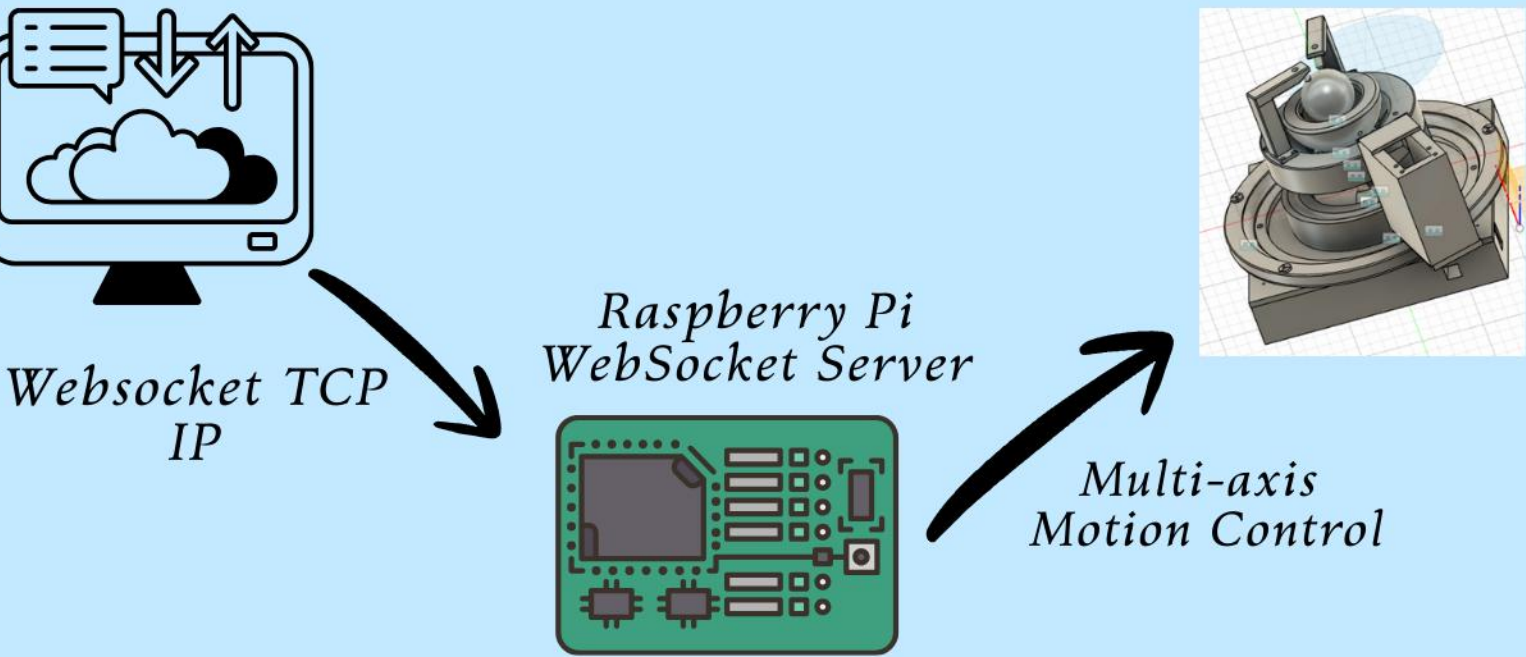
Galipon et al. AIM 2024

Objectives

- Design a simpler multi-axis rotation mechanism using less motors;
- Integrate a hybrid translation + rotation system;
- Automate control using Dynamixel servomotors and an electric linear actuator.
- Develop a complete HMI: 3D visualization, camera, WebSocket, logs.

Methodology

- Design and 3D modeling of the prototype (Fusion 360).
- Electronic integration: Dynamixel, L298N, Raspberry Pi single-board computer.
- Development of an HMI in Python/PyQt6 with 3D visualization.
- Implementation of a WebSocket server.



Basic Principle of OptiSphere V4

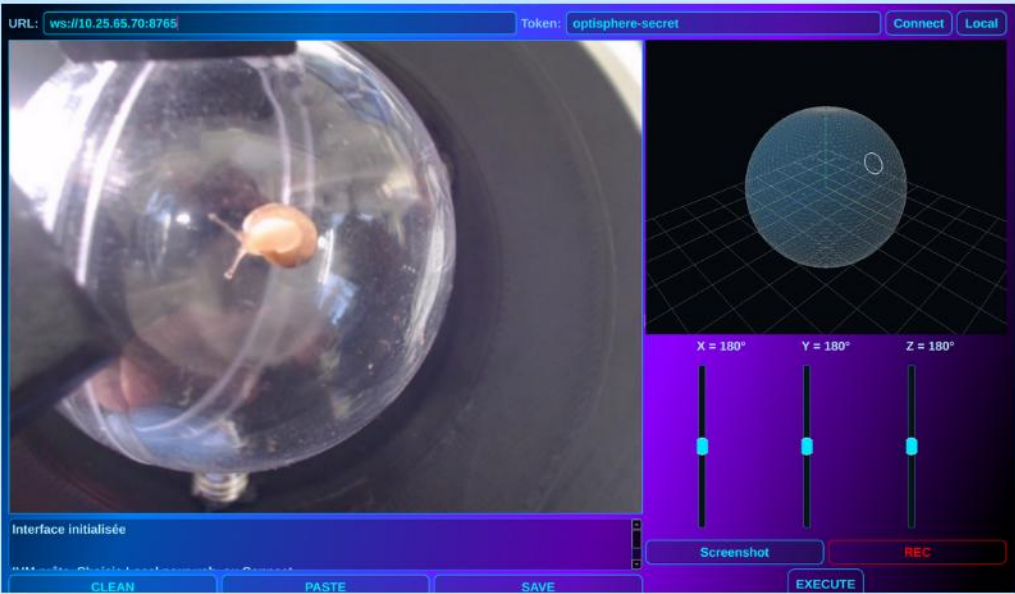
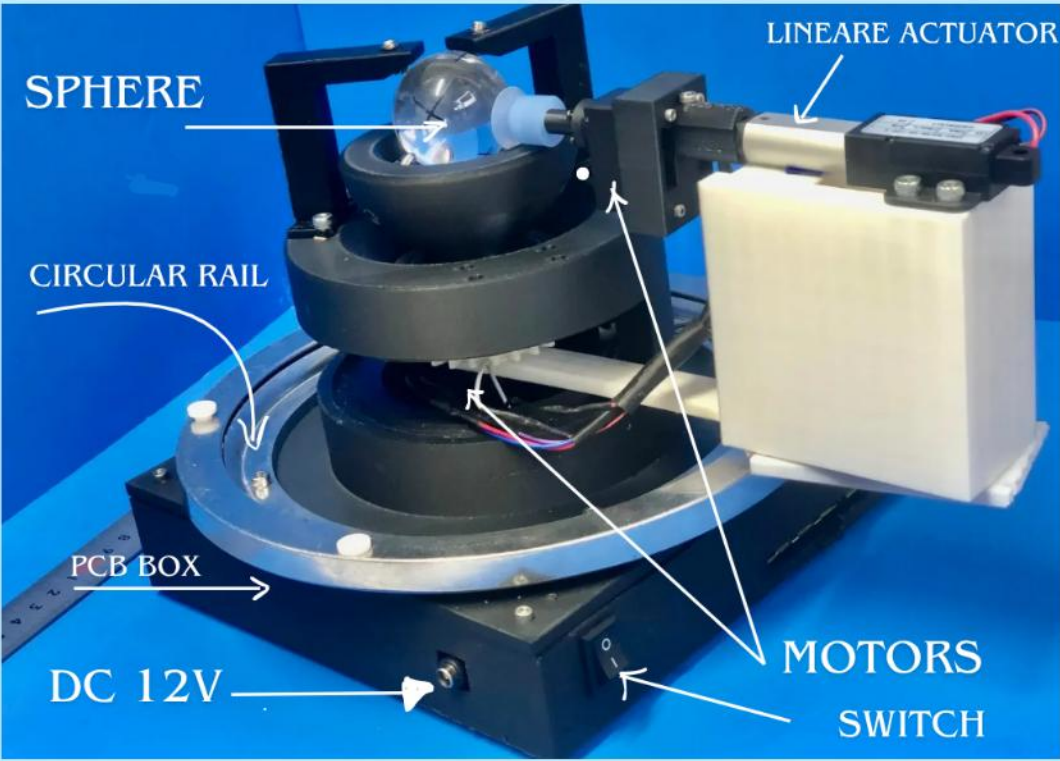
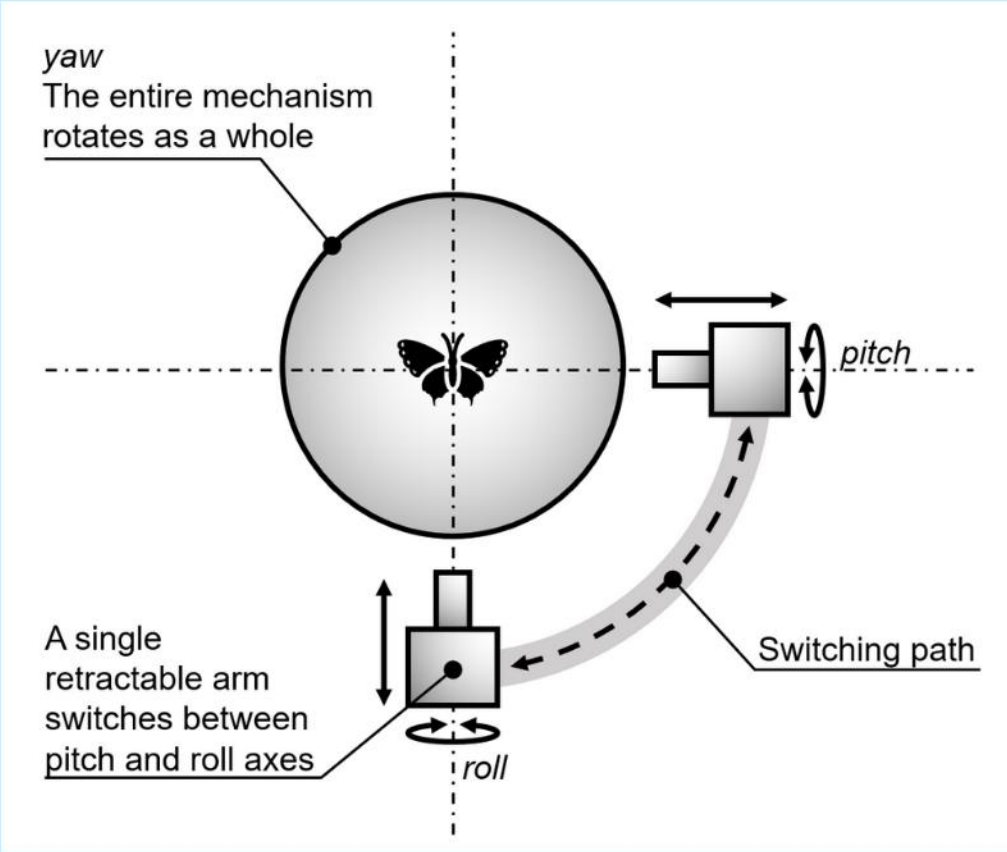
Instead of having two separate retractable arms, a single retractable arm rotates around the sphere to achieve sequential actuation of the pitch and roll axes. The number of caster balls required to fix the sphere was also reduced, making it easier to insert the sphere.

Results

- Controlled rotation along pitch, roll, and yaw.
- Fully operational HMI.

Compared to last version :

- the sphere is much simpler to position
- the number of servomotors and electronic components has been reduced
- the overall platform is more intuitive and user-friendly.

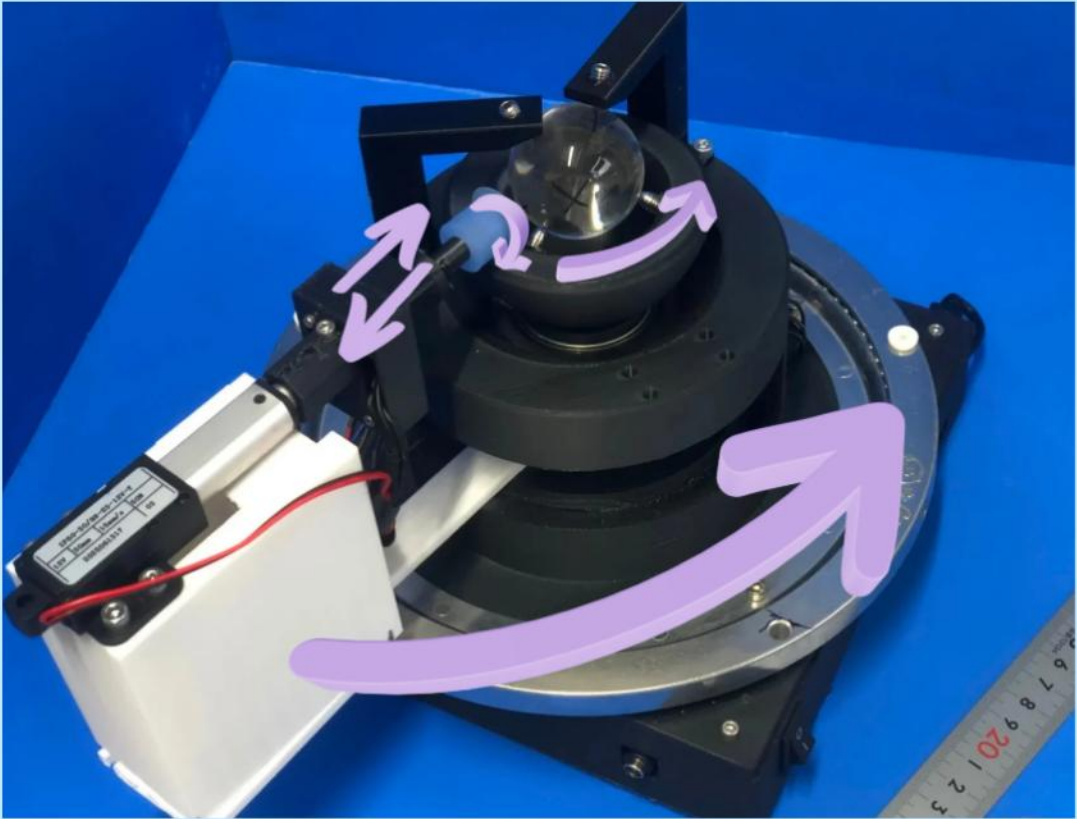


The mechanism was tested by observing a snail crawling on the inside of the sphere, which represents an infinite surface.

Discussion

This version confirms the effectiveness of the upgraded architecture, whose purpose is to simplify the previous system while preserving full multi-axis capability. The hybrid translation-rotation mechanism offers a streamlined and reliable approach for achieving multidirectional 3D observation.

Future work will focus on the software tracking function, by implementing feedback between the camera and the motors.



Conclusion

This work demonstrates the feasibility of a compact multi-axis platform for orienting a spherical sample holder.

The mechanical design, embedded electronics, and software interface together form a functional prototype capable of controlled 3D motion.

The system provides a solid foundation for future developments aimed at improving precision, stability, and integration into advanced microscopy workflows.